

ECON 320 — LECTURE NOTES

Optimisation of Single-Variable Functions

Chiang & Wainwright (2005) Chapters 9–10 | *Hoy et al. (2011)* |
Simon & Blume (1994)

Course ECON 320: Mathematical Economics III

Primary Text A. C. Chiang & K. Wainwright, *Fundamental Methods of Mathematical Economics*, 4th ed., McGraw-Hill, 2005.

Supplementary Hoy, Livernois, McKenna, Rees & Stengos, *Mathematics for Economics*, 3rd ed., MIT Press, 2011.

Simon & Blume, *Mathematics for Economists*, Norton, 1994.

Topics Extreme values, first- and second-order conditions, higher-order conditions, global vs. local optima, concavity and convexity, economic applications.

Date May 2026

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1 Introduction: Why Optimisation?

Optimisation lies at the heart of economic theory. Every canonical economic model involves an agent maximising or minimising something:

- A **consumer** maximises utility subject to a budget constraint.
- A **firm** maximises profit or minimises cost.
- A **social planner** maximises welfare subject to resource constraints.
- A **government** minimises the deadweight loss of taxation.

The single-variable case — where one choice variable determines the objective — is both the simplest setting and the essential building block for multi-variable optimisation. Mastery of this case instils the logic that extends to Lagrangian methods, constrained optimisation, and dynamic programming.

What We Are Looking For

Given an objective function $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$, we seek:

- the value(s) $x^* \in D$ at which f is largest (**maximiser**) or smallest (**minimiser**);
- the **optimal value** $f(x^*)$;
- conditions that *characterise* x^* so we can find it analytically and verify it.

The conditions come in two layers: **first-order conditions** (FOCs), which are necessary, and **second-order conditions** (SOCs), which provide sufficiency.

2 Extreme Values: Maxima and Minima

2.1 Definitions

Definition Local (Relative) and Global (Absolute) Extrema

Let $f : D \rightarrow \mathbb{R}$.

- f has a **local maximum** at x^* if there exists an open interval I containing x^* such that $f(x^*) \geq f(x)$ for all $x \in I \cap D$.
- f has a **local minimum** at x^* if there exists an open interval I containing x^* such that $f(x^*) \leq f(x)$ for all $x \in I \cap D$.
- f has a **global (absolute) maximum** at x^* if $f(x^*) \geq f(x)$ for all $x \in D$.
- f has a **global (absolute) minimum** at x^* if $f(x^*) \leq f(x)$ for all $x \in D$.
- A **strict** local/global extremum replaces $\geq / \leq$ with $> / <$.

The collective term for a maximum or minimum is an **extremum** (pl. **extrema**).

Local vs. Global: A Critical Distinction

A global extremum is always a local extremum, but the converse is *false*. A function can have many local maxima and minima, yet only one global maximum and one global minimum (if they exist). In economic optimisation we almost always seek the *global* optimum, but the calculus conditions we derive initially characterise *local* optima. Additional conditions (concavity/convexity, or checking boundary values) are needed to confirm a local optimum is global.

2.2 The Extreme Value Theorem

Before searching for extrema, it is useful to know whether they exist.

Theorem Extreme Value Theorem (Weierstrass)

If f is **continuous** on a **closed, bounded** interval $[a, b]$, then f attains both its global maximum and its global minimum on $[a, b]$. That is, there exist $x_{\min}, x_{\max} \in [a, b]$ such that:

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}) \quad \text{for all } x \in [a, b].$$

When the EVT Fails

The EVT requires *both* conditions:

- **Continuity:** $f(x) = 1/x$ on $(0, 1]$ is continuous but the supremum $(+\infty)$ is not attained.
- **Closed and bounded domain:** $f(x) = x$ on $(0, 1)$ is continuous but has no maximum (1 not in domain) and no minimum (0 not in domain).

Many economic problems have open or unbounded domains (e.g. $Q > 0$). In such cases we must verify that any critical point found is indeed a maximum — the EVT does not guarantee existence.

2.3 Where Can Extrema Occur?**Theorem Fermat's Theorem**

If f has a local extremum at an interior point x^* of its domain, and if f is differentiable at x^* , then:

$$f'(x^*) = 0$$

Proof. Suppose f has a local maximum at x^* (the minimum case is analogous). By definition, $f(x^* + h) \leq f(x^*)$ for all sufficiently small $|h|$.

For $h > 0$: $\frac{f(x^* + h) - f(x^*)}{h} \leq 0$, so $\lim_{h \rightarrow 0^+} \frac{f(x^* + h) - f(x^*)}{h} \leq 0$.

For $h < 0$: $\frac{f(x^* + h) - f(x^*)}{h} \geq 0$, so $\lim_{h \rightarrow 0^-} \frac{f(x^* + h) - f(x^*)}{h} \geq 0$.

Since f is differentiable, both one-sided limits equal $f'(x^*)$. Therefore $f'(x^*) \leq 0$

and $f'(x^*) \geq 0$ simultaneously, giving $f'(x^*) = 0$. □

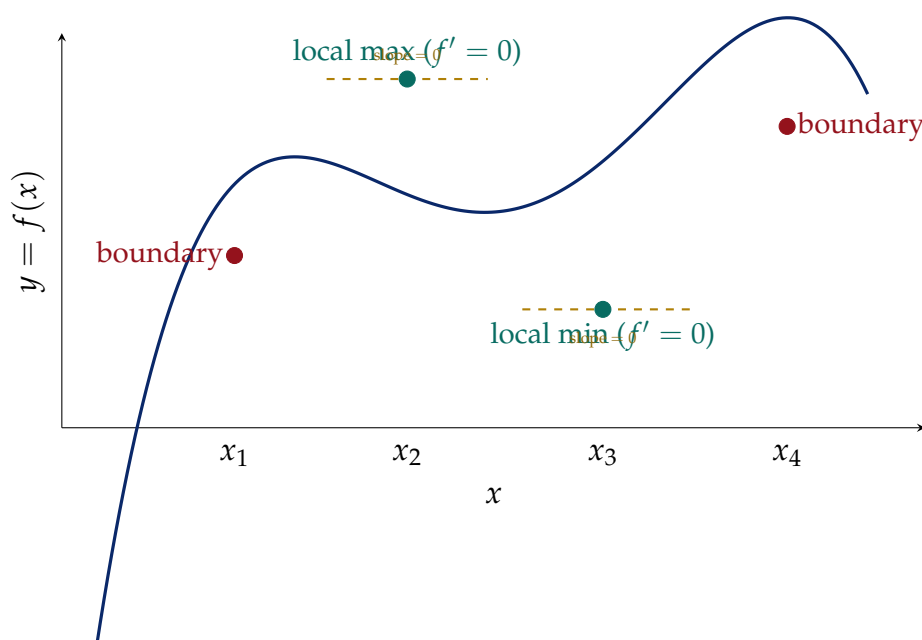
Definition Critical Point (Stationary Point)

A point x^* is a **critical point** (or **stationary point**) of f if $f'(x^*) = 0$. The value $f(x^*)$ is called a **critical value**.

Fermat's Theorem Gives Only Necessary Conditions

- Not every critical point is an extremum. **Example:** $f(x) = x^3$, $f'(0) = 0$ but $x = 0$ is *neither* a local max *nor* a local min (it is an inflection point).
- Extrema can occur at **boundary points** (where f' may not be zero) and at **non-differentiable points** (kinks). **Example:** $f(x) = |x|$ on $[-1, 1]$ has a minimum at $x = 0$ where $f'(0)$ does not exist.

The complete list of *candidates* for extrema on a closed interval $[a, b]$ is: (i) interior critical points, (ii) boundary points a and b , (iii) points where f' does not exist.



Interior critical points (teal) satisfy $f' = 0$. Boundary points (crimson) must also be checked for global extrema.

3 First-Order Conditions (FOC)

3.1 The First-Order Condition

Theorem First-Order Necessary Condition

Let f be differentiable on an open interval containing x^* . A **necessary condition** for x^* to be a local maximum or local minimum of f is:

$$f'(x^*) = 0$$

This is the **first-order condition** (FOC). The economic interpretation is that the marginal effect of x on the objective equals zero at the optimum — the last unit of x neither adds nor subtracts value.

Economic Meaning of the FOC

- **Profit max:** FOC sets marginal revenue equal to marginal cost — $MR(Q^*) = MC(Q^*)$.
- **Utility max** (single good, no constraint): FOC sets marginal utility equal to price.
- **Cost min:** FOC sets the marginal cost of the factor equal to its price.

In every case the FOC expresses an equilibrium between marginal benefit and marginal cost.

Worked Example Finding Critical Points

Find all critical points of $f(x) = 2x^3 - 9x^2 + 12x - 4$.

Step 1 — Differentiate:

$$f'(x) = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x - 1)(x - 2)$$

Step 2 — Set $f'(x) = 0$:

$$6(x - 1)(x - 2) = 0 \implies x_1 = 1, \quad x_2 = 2$$

Step 3 — Compute critical values:

$$f(1) = 2 - 9 + 12 - 4 = 1, \quad f(2) = 16 - 36 + 24 - 4 = 0$$

Both $x = 1$ and $x = 2$ are candidates for extrema. We have not yet determined which is a max, which is a min, or whether either is a local or global extremum — that requires second-order analysis.

3.2 The First Derivative Test

The first derivative test uses the *sign* of f' on either side of a critical point to classify it without computing f'' .

Theorem First Derivative Test

Suppose $f'(x^*) = 0$ and f' is continuous near x^* .

- If f' changes from **positive to negative** as x increases through x^* : x^* is a **local maximum**.
- If f' changes from **negative to positive** as x increases through x^* : x^* is a **local minimum**.
- If f' **does not change sign**: x^* is **neither** a local maximum nor a local minimum (it is an *inflection point*).

Worked Example First Derivative Test Applied

Continue with $f(x) = 2x^3 - 9x^2 + 12x - 4$, $f'(x) = 6(x - 1)(x - 2)$.

Interval	$(x - 1)$	$(x - 2)$	$f'(x)$	f is ...	Conclusion
$x < 1$	—	—	+	increasing	
$x = 1$	0	—	0	stationary	local max: $f(1) = 1$
$1 < x < 2$	+	—	—	decreasing	
$x = 2$	+	0	0	stationary	local min: $f(2) = 0$
$x > 2$	+	+	+	increasing	

The sign of f' changes from + to — at $x = 1$: **local maximum**.

The sign of f' changes from — to + at $x = 2$: **local minimum**.

4 Second-Order Conditions (SOC)

4.1 The Second Derivative and Curvature

The second derivative $f''(x)$ measures how rapidly f' is changing — the *curvature* of f . This curvature determines whether a critical point is a maximum or minimum.

- At a **local maximum**: f' changes from + to —, so f' is *decreasing*. By definition of a decreasing function, $f''(x^*) \leq 0$.
- At a **local minimum**: f' changes from — to +, so f' is *increasing*. By definition, $f''(x^*) \geq 0$.

Theorem Second-Order Sufficient Conditions

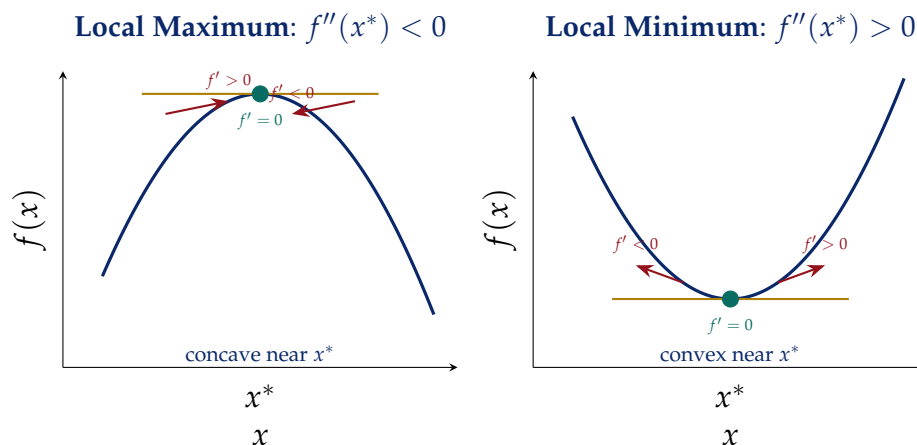
Let f be twice differentiable and suppose $f'(x^*) = 0$.

- If $f''(x^*) < 0$: x^* is a **strict local maximum**.
- If $f''(x^*) > 0$: x^* is a **strict local minimum**.
- If $f''(x^*) = 0$: **inconclusive** (need higher-order conditions — see Section 5).

Reference: Chiang & Wainwright (2005), p. 236; Hoy et al. (2011), p. 286; Simon & Blume (1994), p. 137.

Geometric Intuition for the SOC

- $f''(x^*) < 0$: the tangent line rotates *clockwise* as x increases through x^* . The function is **concave** (hill shaped) near x^* — characteristic of a maximum.
- $f''(x^*) > 0$: the tangent line rotates *counter-clockwise*. The function is **convex** (bowl shaped) near x^* — characteristic of a minimum.



Tangent line (gold) is horizontal at x^* ; curvature determines whether it is a max (left) or min (right).

Worked Example Applying the FOC and SOC

A monopolist faces inverse demand $P = 80 - 3Q$ and total cost $C(Q) = Q^3 - 8Q^2 + 20Q + 10$ for $Q \geq 0$. Find the profit-maximising output and verify it is a maximum.

Step 1 — Write the profit function:

$$\pi(Q) = R(Q) - C(Q) = (80 - 3Q)Q - (Q^3 - 8Q^2 + 20Q + 10)$$

$$\pi(Q) = 80Q - 3Q^2 - Q^3 + 8Q^2 - 20Q - 10 = -Q^3 + 5Q^2 + 60Q - 10$$

Step 2 — FOC ($\pi'(Q) = 0$):

$$\pi'(Q) = -3Q^2 + 10Q + 60 = 0 \implies 3Q^2 - 10Q - 60 = 0$$

$$Q = \frac{10 \pm \sqrt{100 + 720}}{6} = \frac{10 \pm \sqrt{820}}{6} = \frac{10 \pm 28.64}{6}$$

$Q_1 = (10 + 28.64)/6 \approx 6.44$ and $Q_2 = (10 - 28.64)/6 \approx -3.11$ (rejected: $Q \geq 0$).

So $Q^* \approx 6.44$.

Step 3 — SOC:

$$\pi''(Q) = -6Q + 10$$

$$\pi''(6.44) = -6(6.44) + 10 = -38.64 + 10 = -28.64 < 0 \quad \checkmark$$

The negative second derivative confirms $Q^* \approx 6.44$ is a **local maximum**.

Step 4 — Optimal values: $P^* = 80 - 3(6.44) \approx 60.68$, $\pi^* = -(6.44)^3 + 5(6.44)^2 + 60(6.44) - 10 \approx 205.3$.

Worked Example: Minimising Average Cost

Let

$$C(q) = \frac{1}{3}q^3 - 2q^2 + 5q, \quad q > 0.$$

Then

$$AC(q) = \frac{C(q)}{q} = \frac{1}{3}q^2 - 2q + 5.$$

Differentiate:

$$AC'(q) = \frac{2}{3}q - 2.$$

Set the first derivative equal to zero:

$$\frac{2}{3}q - 2 = 0 \implies q^* = 3.$$

Now check the second derivative:

$$AC''(q) = \frac{2}{3} > 0.$$

So $q = 3$ gives a minimum average cost.

The minimum average cost is

$$AC(3) = \frac{1}{3}(9) - 2(3) + 5 = 3 - 6 + 5 = 2.$$

Marginal cost is

$$MC(q) = C'(q) = q^2 - 4q + 5.$$

Evaluating at $q = 3$,

$$MC(3) = 9 - 12 + 5 = 2.$$

Hence

$$MC(3) = AC(3),$$

which confirms the familiar result that average cost is minimised where marginal cost equals average cost.

4.2 The Relationship Between AC and MC

Proposition AC Minimum and the MC = AC Condition

Let $C(q)$ be a differentiable total cost function with $AC(q) = C(q)/q$. The average cost is minimised where **marginal cost equals average cost**:

$$AC'(q) = 0 \iff MC(q) = AC(q)$$

Proof. $AC(q) = C(q)/q$. Differentiate using the quotient rule:

$$AC'(q) = \frac{C'(q) \cdot q - C(q) \cdot 1}{q^2} = \frac{C'(q) - C(q)/q}{q} = \frac{MC(q) - AC(q)}{q}$$

Setting $AC'(q) = 0$ gives $MC(q) = AC(q)$ (since $q > 0$). □ □

5 Higher-Order Conditions: Resolving the Inconclusive Case

5.1 When $f''(x^*) = 0$

The SOC is inconclusive when $f''(x^*) = 0$. To resolve the classification we examine higher derivatives.

Theorem Higher-Order Derivative Test

Suppose $f'(x^*) = 0$ and $f'', f''', \dots, f^{(n-1)}$ all equal zero at x^* , but $f^{(n)}(x^*) \neq 0$.

- If n is **odd**: x^* is **neither** a local maximum nor a local minimum (it is an inflection point of f , a horizontal inflection).
- If n is **even** and $f^{(n)}(x^*) < 0$: x^* is a **strict local maximum**.
- If n is **even** and $f^{(n)}(x^*) > 0$: x^* is a **strict local minimum**.

Reference: Chiang & Wainwright (2005), pp. 240–242.

The Intuition Behind the Higher-Order Test

The Taylor expansion of f about x^* gives:

$$f(x^* + h) - f(x^*) \approx \frac{f^{(n)}(x^*)}{n!} h^n \quad \text{for small } h$$

(since $f' = \dots = f^{(n-1)} = 0$ at x^*).

- If n is **odd**: h^n changes sign as h crosses zero, so $f(x^* + h) - f(x^*)$ changes sign — x^* is neither a max nor min.
- If n is **even**: $h^n > 0$ for all $h \neq 0$. The sign of $f(x^* + h) - f(x^*)$ is determined solely by $\text{sgn}(f^{(n)}(x^*))$. Negative $\Rightarrow$ max; positive $\Rightarrow$ min.

Worked Example Resolving the Inconclusive SOC: Three Cases

Case A — Horizontal inflection: $f(x) = x^3$.

$$f'(x) = 3x^2, \quad f'(0) = 0. \quad f''(x) = 6x, \quad f''(0) = 0. \quad f'''(0) = 6 \neq 0.$$

Here $n = 3$ (odd): $x = 0$ is a **horizontal inflection point** — neither a max nor a min. Verify: $f(0.1) = 0.001 > 0 = f(0) > f(-0.1) = -0.001$.

Case B — Local minimum with flat second derivative: $f(x) = x^4$.

$$f'(0) = 0, \quad f''(0) = 0, \quad f'''(0) = 0, \quad f^{(4)}(0) = 24 > 0.$$

$n = 4$ (even), $f^{(4)}(0) > 0$: $x = 0$ is a **strict local minimum**. Verify: $f(h) = h^4 > 0 = f(0)$ for all $h \neq 0$. ✓

Case C — Local maximum with flat second derivative: $f(x) = -x^4$.

$$f^{(4)}(0) = -24 < 0.$$

$n = 4$ (even), $f^{(4)}(0) < 0$: $x = 0$ is a **strict local maximum**.

5.2 Summary: Full Classification Procedure

Step	Action
1. Find critical points	Solve $f'(x^*) = 0$.
2. Apply SOC	Compute $f''(x^*)$.
3. If $f''(x^*) < 0$	Local maximum.
4. If $f''(x^*) > 0$	Local minimum.
5. If $f''(x^*) = 0$	Inconclusive; find lowest n with $f^{(n)}(x^*) \neq 0$.
6. n odd	Inflection point (neither max nor min).
7. n even, $f^{(n)} < 0$	Local maximum.
8. n even, $f^{(n)} > 0$	Local minimum.
9. Check boundaries	Evaluate f at domain endpoints (if applicable).
10. Global optimum	Compare all local optima and boundary values.

6 Concavity, Convexity, and Global Optima

6.1 Concavity and Convexity

Definition Concave and Convex Functions

A function $f : I \rightarrow \mathbb{R}$ on an interval I is:

- **Concave** on I if for all $x_1, x_2 \in I$ and $\lambda \in [0, 1]$:

$$f(\lambda x_1 + (1 - \lambda)x_2) \geq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

The chord lies at or below the curve.

- **Convex** on I if for all $x_1, x_2 \in I$ and $\lambda \in [0, 1]$:

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

The chord lies at or above the curve.

- **Strictly concave / convex** if the inequality is strict for $\lambda \in (0, 1)$ and $x_1 \neq x_2$.

Theorem Second-Derivative Characterisation of Concavity

If f is twice differentiable on an open interval I :

- f is **concave** on $I \iff f''(x) \leq 0$ for all $x \in I$.
- f is **strictly concave** on I if $f''(x) < 0$ for all $x \in I$.
- f is **convex** on $I \iff f''(x) \geq 0$ for all $x \in I$.
- f is **strictly convex** on I if $f''(x) > 0$ for all $x \in I$.

Note: $f''(x) < 0$ everywhere implies strictly concave, but strictly concave does not require $f'' < 0$ everywhere (e.g. $f'' = 0$ at an isolated point is fine).

6.2 Global Optima via Concavity/Convexity

The key power of concavity and convexity in optimisation is that they convert local optima into *global* optima.

Theorem Global Optimality from Concavity/Convexity

- If f is **concave** on I and $f'(x^*) = 0$ for some interior $x^* \in I$, then x^* is a **global maximum** of f on I .
- If f is **convex** on I and $f'(x^*) = 0$ for some interior $x^* \in I$, then x^* is a **global minimum** of f on I .
- If f is **strictly concave** on I , there is **at most one** critical point, and it is the unique global maximum.
- If f is **strictly convex** on I , there is **at most one** critical point, and it is the unique global minimum.

Reference: Chiang & Wainwright (2005), p. 248; Simon & Blume (1994), Theorem 3.4.

Proof for the concave case. Suppose f is concave and $f'(x^*) = 0$. For any $x \in I$, take $\lambda \in (0, 1)$ and write $x^* = \lambda x^* + (1 - \lambda)x^*$. By concavity:

$$f(x^*) = f(\lambda x^* + (1 - \lambda)x^*) \geq \lambda f(x^*) + (1 - \lambda)f(x) \quad \text{when } x^* = \lambda x^* + (1 - \lambda)x$$

More directly: for concave f and any x , $f(x) \leq f(x^*) + f'(x^*)(x - x^*) = f(x^*) + 0 = f(x^*)$. (This uses the supporting hyperplane property of concave functions: a concave function lies below any tangent line.) $\square$ $\square$

Why This Matters in Economics

- **Profit max:** if $\pi(Q) = R(Q) - C(Q)$ is globally concave in Q (which holds when R is concave and C is convex), then any critical point Q^* is the unique global profit maximiser.
- **Utility max:** with a strictly concave utility function, any interior solution to the FOC is the unique global maximum.
- **Cost min:** with a strictly convex cost function, the FOC identifies the unique global cost minimiser.
- In practice, many economic objectives are concave (revenue concave, cost convex) by assumption, which is why the FOC alone often suffices in economic models.

Worked Example Global Maximum via Concavity

A firm's revenue is $R(Q) = 120Q - 2Q^2$ and cost is $C(Q) = Q^2 + 20Q + 50$. Show that the profit-maximising output is a global maximum.

Profit: $\pi(Q) = R(Q) - C(Q) = 120Q - 2Q^2 - Q^2 - 20Q - 50 = -3Q^2 + 100Q - 50$.

Global concavity: $\pi''(Q) = -6 < 0$ for all Q . Therefore π is strictly concave globally.

FOC: $\pi'(Q) = -6Q + 100 = 0 \Rightarrow Q^* = 100/6 \approx 16.67$.

Since π is strictly concave, Q^* is the **unique global maximum**. $\pi^* = -3(100/6)^2 + 100(100/6) - 50 = -3(10000/36) + 10000/6 - 50 = -10000/12 + 10000/6 - 50 = 10000/12 - 50 \approx 783.33$.

6.3 Inflection Points

Definition Inflection Point

A point x_0 is an **inflection point** of f if f changes concavity at x_0 :

- f'' changes sign at x_0 (from $+$ to $-$ or from $-$ to $+$).
- Necessarily $f''(x_0) = 0$ if f'' is continuous there.

An inflection point is not an extremum (unless it is also a critical point, making it a horizontal inflection).

Common Error: $f'' = 0$ Does Not Imply Inflection

$f''(x_0) = 0$ is *necessary but not sufficient* for an inflection point.

Example: $f(x) = x^4$. $f''(0) = 0$, but $f'' = 12x^2 \geq 0$ everywhere — f is convex everywhere; $x = 0$ is a minimum, not an inflection point.

Correct test: Verify that f'' changes sign at x_0 .

Worked Example Finding Inflection Points in a Cost Function

$$C(Q) = Q^3 - 9Q^2 + 30Q + 5.$$

$$C'(Q) = 3Q^2 - 18Q + 30 = 3(Q^2 - 6Q + 10)$$

$$C''(Q) = 6Q - 18 = 6(Q - 3)$$

$$C''(Q) = 0 \Rightarrow Q = 3.$$

For $Q < 3$: $C'' < 0$ (MC is falling — economies of scale). For $Q > 3$: $C'' > 0$ (MC is rising — diseconomies of scale).

C'' changes sign at $Q = 3$: this is an **inflection point**. Economically, $Q = 3$ is the *minimum efficient scale* — the output at which MC is lowest and the firm achieves its greatest internal economies of scale.

$$MC(3) = 3(9) - 18(3) + 30 = 27 - 54 + 30 = 3.$$

7 Optimisation on a Closed Interval

When the domain is a **closed, bounded** interval $[a, b]$, the EVT guarantees existence of global extrema. The global maximum (minimum) is the largest (smallest) value among all interior critical points *and* the two boundary points.

Algorithm for Optimisation on $[a, b]$

1. Find all interior critical points: solve $f'(x) = 0$ for $x \in (a, b)$.
2. Evaluate f at every critical point.
3. Evaluate f at the boundary points: $f(a)$ and $f(b)$.
4. The **global maximum** is $\max\{f(a), f(b), f(x_1^*), f(x_2^*), \dots\}$.
5. The **global minimum** is $\min\{f(a), f(b), f(x_1^*), f(x_2^*), \dots\}$.

Worked Example Global Optimum on a Closed Interval

Find the global maximum and minimum of $f(x) = x^3 - 3x^2 - 9x + 5$ on $[-2, 6]$.

Step 1 — Find critical points:

$$f'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x + 1)(x - 3)$$

$$f'(x) = 0 \Rightarrow x = -1 \text{ and } x = 3. \text{ Both lie in } (-2, 6).$$

Step 2 — Evaluate at critical points:

$$f(-1) = -1 - 3 + 9 + 5 = 10, \quad f(3) = 27 - 27 - 27 + 5 = -22$$

Step 3 — Evaluate at boundaries:

$$f(-2) = -8 - 12 + 18 + 5 = 3, \quad f(6) = 216 - 108 - 54 + 5 = 59$$

Step 4 — Compare all values:

Point	x	$f(x)$
Left boundary	-2	3
Critical point (local max)	-1	10
Critical point (local min)	3	-22
Right boundary	6	59

Global maximum: $f(6) = 59$ (at boundary).

Global minimum: $f(3) = -22$ (at interior critical point).

Key lesson: The local maximum at $x = -1$ with $f(-1) = 10$ is *not* the global maximum — the boundary value $f(6) = 59$ is larger. Always check boundaries when the domain is closed.

8 Monotone Functions and Optimisation

Definition Monotone Transformations

A **strictly increasing transformation** g (i.e. $g' > 0$) preserves the location of optima:

$$x^* = \arg \max_x f(x) \iff x^* = \arg \max_x g(f(x))$$

This result is used constantly in economics: we maximise $\ln U$ instead of U (since $\ln$ is strictly increasing) to simplify computation.

Worked Example Log Transformation in Cobb-Douglas Maximisation

A consumer's utility is $U = x_1^\alpha x_2^{1-\alpha}$ (with only one good for illustration: $U = x^\alpha$, $\alpha \in (0, 1)$, $x \geq 0$).

Direct approach: $U'(x) = \alpha x^{\alpha-1} > 0$ for all $x > 0$: utility is strictly increasing, so it is maximised at the budget constraint boundary (no interior maximum in one good).

Two-good case illustration: Consider the utility maximisation $\max_L \ln U$ where $U = L^{0.4}(T - L)^{0.6}$ and T is total time.

$$\ln U = 0.4 \ln L + 0.6 \ln(T - L)$$

$$\frac{d \ln U}{dL} = \frac{0.4}{L} - \frac{0.6}{T-L} = 0$$

$$\Rightarrow 0.4(T - L) = 0.6L \Rightarrow 0.4T = L(0.6 + 0.4) = L$$

$$\Rightarrow L^* = 0.4T$$

This is the time-allocation optimum. Using $\ln U$ instead of U turns a rational-power maximisation into a simple linear equation in L .

9 Economic Applications

9.1 Tax Incidence and Revenue Maximisation

Worked Example Government Revenue Maximisation (Laffer Curve)

Suppose the government levies a proportional income tax at rate $t \in [0, 1]$. Taxable income is $Y(t) = Y_0(1 - t)^\gamma$ where $\gamma > 0$ captures the labour-supply elasticity ($Y_0 > 0$ is pre-tax potential income).

Tax revenue:

$$R(t) = t \cdot Y(t) = t Y_0(1 - t)^\gamma$$

FOC: $R'(t) = 0$:

$$R'(t) = Y_0(1 - t)^\gamma + t Y_0 \gamma (1 - t)^{\gamma-1}(-1) = Y_0(1 - t)^{\gamma-1}[(1 - t) - \gamma t] = Y_0(1 - t)^{\gamma-1}[1 - t(1 + \gamma)]$$

Set $R'(t) = 0$: either $t = 1$ (trivial) or

$$t^* = \frac{1}{1 + \gamma}$$

SOC: It can be verified that $R''(t^*) < 0$, so t^* is a maximum.

Interpretation:

- If $\gamma = 0$ (inelastic supply): $t^* = 1$ — revenue is maximised at a 100% tax rate (income does not respond).
- If $\gamma = 1$: $t^* = 1/2$ — the revenue-maximising rate is 50%.
- If $\gamma = 2$: $t^* = 1/3$ — elastic supply pushes the optimal rate lower.

This is the **Laffer curve**: $R(0) = 0$, $R(1) = 0$, $R(t^*) > 0$.

9.2 Price Discrimination

Worked Example Third-Degree Price Discrimination

A monopolist sells in two markets with inverse demands $P_1 = 24 - Q_1$ and $P_2 = 16 - Q_2/2$. Marginal cost is constant at $MC = 4$.

Profit:

$$\pi = (P_1 - 4)Q_1 + (P_2 - 4)Q_2 = (20 - Q_1)Q_1 + (12 - Q_2/2)Q_2$$

Since the two markets are independent, maximise separately.

Market 1:

$$\pi_1(Q_1) = 20Q_1 - Q_1^2 \Rightarrow \pi_1' = 20 - 2Q_1 = 0 \Rightarrow Q_1^* = 10, P_1^* = 14$$

$$\pi_1'' = -2 < 0: \text{maximum. } \checkmark$$

Market 2:

$$\pi_2(Q_2) = 12Q_2 - Q_2^2/2 \Rightarrow \pi_2' = 12 - Q_2 = 0 \Rightarrow Q_2^* = 12, P_2^* = 10$$

$$\pi_2'' = -1 < 0: \text{maximum. } \checkmark$$

Total profit: $\pi^* = (14 - 4)(10) + (10 - 4)(12) = 100 + 72 = 172.$

Elasticity check: $\varepsilon_1 = \frac{dQ_1}{dP_1} \cdot \frac{P_1}{Q_1} = (-1) \cdot \frac{14}{10} = -1.4$

$\varepsilon_2 = \frac{dQ_2}{dP_2} \cdot \frac{P_2}{Q_2} = (-2) \cdot \frac{10}{12} = -1.67$

The market with less elastic demand ($|\varepsilon_1| < |\varepsilon_2|$) faces the higher price ($P_1 > P_2$), consistent with the inverse elasticity rule.

9.3 Optimal Resource Depletion

Worked Example Hotelling's Rule — Optimal Extraction

A mine owner has a stock S_0 of a non-renewable resource. Extraction $q(t)$ over time T yields revenue $P(q)q$ and costs $C(q)$. The static net benefit function (profit) in period t is:

$$\pi(q) = P(q)q - C(q)$$

The owner maximises the present value of profits. For a fixed period t , the optimal q_t^* satisfies:

$$P'(q^*)q^* + P(q^*) - C'(q^*) = 0 \iff \text{MR}(q^*) = \text{MC}(q^*)$$

Let λ_t be the shadow price (royalty) of the resource (the opportunity cost of extracting one more unit today instead of tomorrow). The full profit-maximising condition becomes:

$$\text{MR}(q^*) = \text{MC}(q^*) + \lambda_t$$

Hotelling's Rule states that the net price $P(q^*) - \text{MC}(q^*)$ must grow at the rate of interest r : $\dot{\lambda}/\lambda = r$ — in equilibrium, the scarcity rent rises at the interest rate, making the owner indifferent between extraction now and later.

10 Practice Questions

10.1 Section A: Extreme Values and Classification

A1. (Critical Points and Classification) For each function, find all critical points and classify them as local maximum, local minimum, or inflection point using (i) the first derivative test and (ii) the second derivative test (where applicable).

(a) $f(x) = x^3 - 6x^2 + 9x + 2$

(b) $g(x) = \frac{1}{4}x^4 - \frac{2}{3}x^3 - 3x^2 + 4$

(c) $h(x) = xe^{-x}$ for $x \in \mathbb{R}$

(d) $k(x) = x^2 \ln x$ for $x > 0$

A2. (Higher-Order Conditions) Classify the critical point $x = 0$ for each function using the higher-order derivative test.

(a) $f(x) = 1 - x^6$

(b) $g(x) = x^5$

(c) $h(x) = x^4 - 2x^2$ (find all critical points first)

A3. (Closed Interval Optimisation) Find the global maximum and minimum of each function on the given interval.

(a) $f(x) = x^3 - 3x + 1$ on $[-2, 3]$

(b) $g(x) = \frac{x}{x^2 + 1}$ on $[-3, 3]$

(c) $h(x) = e^x - 3x$ on $[0, 3]$

A4. (Concavity and Global Optima)

(a) Show that $f(x) = -2x^2 + 8x - 3$ is strictly concave and find its global maximum.

(b) Show that $g(x) = e^x + e^{-x}$ is strictly convex and find its global minimum.

(c) A function f is concave on $\mathbb{R}$ and $f'(5) = 0$. Without further calculation, what can you conclude about $x = 5$?

A5. (Inflection Points) Find all inflection points of $f(x) = x^4 - 4x^3 + 6x^2 - 4x + 1$. Determine the intervals of concavity and convexity.

10.2 Section B: Economic Applications

- B1. (Monopoly Output and Welfare)** A monopolist faces demand $P(Q) = 100 - Q$ and has total cost $C(Q) = \frac{1}{4}Q^2 + 10Q + 100$.
- (a) Find the profit-maximising output Q^* , price P^* , and maximum profit π^* .
 - (b) Verify the SOC and show that π is globally concave.
 - (c) Compute the price elasticity of demand at Q^* .
 - (d) Using the result $MR = P(1 + 1/\varepsilon)$, verify that $MR = MC$ at Q^* .
 - (e) Find the competitive output Q_c where $P = MC$. Compute the deadweight loss from monopoly.
- B2. (Average and Marginal Cost)** A firm's total cost function is $C(Q) = Q^3 - 12Q^2 + 60Q + 100$.
- (a) Derive expressions for MC, AC, and AVC.
 - (b) Find the output that minimises MC. What type of critical point is this for $C(Q)$?
 - (c) Find the output that minimises AC and verify that $AC = MC$ at this point.
 - (d) Find the output that minimises AVC.
 - (e) On what interval is the cost function convex? What is the economic interpretation of this interval?
- B3. (Revenue Maximisation)** A firm faces demand $Q = 60 - 2P + 0.5P^2$ (non-linear demand).
- (a) Express total revenue R as a function of price P .
 - (b) Find the revenue-maximising price.
 - (c) Show that at the revenue-maximising price, demand elasticity equals -1 (unit elastic).
 - (d) Is the revenue function globally concave in P ?
- B4. (Optimal Advertising)** A monopolist's profit is $\pi = R(Q) - C(Q) - A$, where A is advertising expenditure. Demand depends on advertising: $Q(A) = 20\sqrt{A}$. Revenue is $R = 50Q$ and cost is $C = 10Q$.
- (a) Write profit as a function of A alone.
 - (b) Find the profit-maximising advertising level A^* .
 - (c) Verify the SOC.
 - (d) Compute the *advertising elasticity of profit*: how does a 1% increase in A affect π ?

B5. (Tax Policy — Government Revenue) The government imposes a per-unit tax t on a good. The competitive market clears at: $Q^*(t) = \max(0, 100 - 2t)$. Tax revenue is $T(t) = t \cdot Q^*(t)$.

- (a) Find the tax rate t^* that maximises tax revenue.
- (b) What is the maximum tax revenue?
- (c) Show that $T''(t^*) < 0$.
- (d) For $t > t^*$, what happens to revenue when t increases? Relate this to the Laffer curve.
- (e) At what tax rate does the market shut down ($Q^* = 0$)?

11 Full Solutions

11.1 Solutions: Section A

Solution

A1. Critical Points and Classification

(a) $f(x) = x^3 - 6x^2 + 9x + 2$

$$f'(x) = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3)$$

Critical points: $x = 1$ and $x = 3$.

First derivative test:

Interval	$(x - 1)$	$(x - 3)$	f'	Behaviour
$x < 1$	—	—	+	increasing
$x = 1$	0	—	0	stationary
$1 < x < 3$	+	—	—	decreasing
$x = 3$	+	0	0	stationary
$x > 3$	+	+	+	increasing

$x = 1$: f' goes $+$ $\rightarrow$ $-$: **local maximum**, $f(1) = 1 - 6 + 9 + 2 = 6$.

$x = 3$: f' goes $-$ $\rightarrow$ $+$: **local minimum**, $f(3) = 27 - 54 + 27 + 2 = 2$.

Second derivative test: $f''(x) = 6x - 12$.

$f''(1) = 6 - 12 = -6 < 0$: local max. $\checkmark$ $f''(3) = 18 - 12 = 6 > 0$: local min. $\checkmark$

(b) $g(x) = \frac{1}{4}x^4 - \frac{2}{3}x^3 - 3x^2 + 4$

$$g'(x) = x^3 - 2x^2 - 6x = x(x^2 - 2x - 6) = x(x - (1 + \sqrt{7}))(x - (1 - \sqrt{7}))$$

Critical points: $x = 0$, $x = 1 + \sqrt{7} \approx 3.646$, $x = 1 - \sqrt{7} \approx -1.646$.

$$g''(x) = 3x^2 - 4x - 6.$$

$g''(0) = -6 < 0$: **local maximum**, $g(0) = 4$.

$g''(3.646) = 3(13.29) - 14.58 - 6 = 39.88 - 20.58 = 19.3 > 0$: **local minimum**.

$g''(-1.646) = 3(2.71) + 6.58 - 6 = 8.13 + 0.58 = 8.71 > 0$: **local minimum**.

(c) $h(x) = xe^{-x}$

$$h'(x) = e^{-x} + x(-1)e^{-x} = e^{-x}(1 - x)$$

$$h'(x) = 0 \Rightarrow x = 1 \text{ (since } e^{-x} > 0 \text{ always).}$$

$$h''(x) = -e^{-x}(1 - x) + e^{-x}(-1) = e^{-x}(x - 2)$$

$$h''(1) = e^{-1}(1 - 2) = -e^{-1} < 0$$
: **local (and global) maximum**, $h(1) = e^{-1}$.

Globally: $h(x) \rightarrow 0$ as $x \rightarrow \infty$, $h(x) \rightarrow -\infty$ as $x \rightarrow -\infty$, and h is strictly concave near $x = 1$ — the only critical point is the global max over all $x \geq 0$.

(d) $k(x) = x^2 \ln x$, $x > 0$

$$k'(x) = 2x \ln x + x^2 \cdot \frac{1}{x} = 2x \ln x + x = x(2 \ln x + 1)$$

$$k'(x) = 0 \Rightarrow x = 0 \text{ (excluded) or } 2 \ln x + 1 = 0 \Rightarrow \ln x = -1/2 \Rightarrow x^* = e^{-1/2} = 1/\sqrt{e}.$$

$$k''(x) = 2 \ln x + 1 + 2x \cdot \frac{1}{x} = 2 \ln x + 3$$

$$k''(1/\sqrt{e}) = 2(-1/2) + 3 = -1 + 3 = 2 > 0$$
: **local minimum**.

$$k(1/\sqrt{e}) = (1/\sqrt{e})^2 \ln(1/\sqrt{e}) = \frac{1}{e} \cdot \left(-\frac{1}{2}\right) = -\frac{1}{2e}.$$

Solution**A2. Higher-Order Conditions**

(a) $f(x) = 1 - x^6$. $f'(0) = 0$.

$f''(x) = -30x^4$, $f''(0) = 0$. $f'''(0) = 0$, $f^{(4)}(0) = 0$, $f^{(5)}(0) = 0$, $f^{(6)}(x) = -720$, so $f^{(6)}(0) = -720 < 0$.

$n = 6$ (even), negative: $x = 0$ is a **strict local maximum**.

Confirm: $f(h) = 1 - h^6 < 1 = f(0)$ for $h \neq 0$. ✓

(b) $g(x) = x^5$. $g'(0) = 0$.

$g''(0) = 0$, $g'''(0) = 0$, $g^{(4)}(0) = 0$, $g^{(5)}(0) = 120 \neq 0$.

$n = 5$ (odd): $x = 0$ is a **horizontal inflection point**.

Confirm: $g(h) = h^5 > 0$ for $h > 0$ and $g(h) < 0$ for $h < 0$, so g crosses zero at $x = 0$ — neither max nor min. ✓

(c) $h(x) = x^4 - 2x^2$.

$h'(x) = 4x^3 - 4x = 4x(x^2 - 1) = 4x(x - 1)(x + 1)$

Critical points: $x = 0$, $x = 1$, $x = -1$.

$h''(x) = 12x^2 - 4$.

$h''(0) = -4 < 0$: $x = 0$ is a **local maximum**, $h(0) = 0$.

$h''(1) = 12 - 4 = 8 > 0$: $x = 1$ is a **local minimum**, $h(1) = -1$.

$h''(-1) = 8 > 0$: $x = -1$ is a **local minimum**, $h(-1) = -1$.

Note: The question asks about $x = 0$ specifically. SOC already resolves it ($h''(0) < 0$), so the higher-order test is not needed.

Solution**A3. Closed Interval Optimisation**

(a) $f(x) = x^3 - 3x + 1$ on $[-2, 3]$.

$f'(x) = 3x^2 - 3 = 3(x^2 - 1) = 3(x - 1)(x + 1)$. Critical points: $x = -1, 1 \in (-2, 3)$.

Evaluate:

x	$f(x)$
-2	$-8 + 6 + 1 = -1$
-1	$-1 + 3 + 1 = 3$
1	$1 - 3 + 1 = -1$
3	$27 - 9 + 1 = 19$

Global max: $f(3) = 19$ (right boundary). **Global min:** $f(-2) = f(1) = -1$ (tied).

(b) $g(x) = \frac{x}{x^2 + 1}$ on $[-3, 3]$.

$g'(x) = \frac{(x^2 + 1) - x(2x)}{(x^2 + 1)^2} = \frac{1 - x^2}{(x^2 + 1)^2}$

$g'(x) = 0 \Rightarrow x = \pm 1$.

Evaluate:

x	$g(x)$
-3	$-3/10 = -0.3$
-1	$-1/2 = -0.5$
1	$1/2 = 0.5$
3	$3/10 = 0.3$

Global max: $g(1) = 1/2$. **Global min:** $g(-1) = -1/2$.(c) $h(x) = e^x - 3x$ on $[0, 3]$. $h'(x) = e^x - 3 = 0 \Rightarrow x = \ln 3 \approx 1.099 \in (0, 3)$.

Evaluate:

x	$h(x)$
0	1
$\ln 3$	$3 - 3 \ln 3 \approx 3 - 3.296 = -0.296$
3	$e^3 - 9 \approx 20.09 - 9 = 11.09$

Global max: $h(3) = e^3 - 9 \approx 11.09$. **Global min:** $h(\ln 3) = 3 - 3 \ln 3 \approx -0.296$.**Solution****A4. Concavity and Global Optima**(a) $f(x) = -2x^2 + 8x - 3$. $f''(x) = -4 < 0$ for all $x \in \mathbb{R}$: **strictly concave globally**.FOC: $f'(x) = -4x + 8 = 0 \Rightarrow x^* = 2$.Since f is strictly concave, $x^* = 2$ is the **unique global maximum**: $f(2) = -8 + 16 - 3 = 5$.(b) $g(x) = e^x + e^{-x}$. $g''(x) = e^x + e^{-x} > 0$ for all $x \in \mathbb{R}$: **strictly convex globally**.FOC: $g'(x) = e^x - e^{-x} = 0 \Rightarrow e^{2x} = 1 \Rightarrow x^* = 0$.Since g is strictly convex, $x^* = 0$ is the **unique global minimum**: $g(0) = 1 + 1 = 2$.(c) Since f is concave on $\mathbb{R}$ and $f'(5) = 0$, the supporting hyperplane property of concave functions gives:

$$f(x) \leq f(5) + f'(5)(x - 5) = f(5) + 0 = f(5) \quad \text{for all } x \in \mathbb{R}.$$

Therefore $x = 5$ is a **global maximum** of f on $\mathbb{R}$. No further calculation is needed.**Solution****A5. Inflection Points**

$$f(x) = x^4 - 4x^3 + 6x^2 - 4x + 1 = (x - 1)^4$$

$$f'(x) = 4(x - 1)^3, \quad f''(x) = 12(x - 1)^2$$

$$f''(x) = 0 \Rightarrow x = 1.$$

Check sign change: $f''(x) = 12(x - 1)^2 \geq 0$ for all x , with equality only at $x = 1$.

The sign of f'' does *not* change at $x = 1$ — it is non-negative on both sides.

Therefore $x = 1$ is **not an inflection point**.

$f(x) = (x - 1)^4 \geq 0$ for all x , with minimum value $f(1) = 0$: $x = 1$ is a **global minimum**, not an inflection point.

Intervals of concavity/convexity: $f''(x) = 12(x - 1)^2 \geq 0$ everywhere. f is **weakly convex on all of $\mathbb{R}$** (strictly convex except at $x = 1$).

There are **no inflection points**.

Key lesson: $f'' = 0$ at a point does not imply inflection. The sign of f'' must actually change.

11.2 Solutions: Section B

Solution

B1. Monopoly Output and Welfare

$$P(Q) = 100 - Q, \quad C(Q) = \frac{1}{4}Q^2 + 10Q + 100.$$

$$(a) \pi(Q) = (100 - Q)Q - \frac{1}{4}Q^2 - 10Q - 100 = 100Q - Q^2 - \frac{1}{4}Q^2 - 10Q - 100 = -\frac{5}{4}Q^2 + 90Q - 100$$

$$\text{FOC: } \pi'(Q) = -\frac{5}{2}Q + 90 = 0 \Rightarrow Q^* = 36.$$

$$P^* = 100 - 36 = 64.$$

$$\pi^* = -\frac{5}{4}(36)^2 + 90(36) - 100 = -\frac{5}{4}(1296) + 3240 - 100 = -1620 + 3240 - 100 = 1520.$$

(b) SOC: $\pi''(Q) = -\frac{5}{2} < 0$ for all Q : π is **strictly concave globally** and $Q^* = 36$ is the unique global maximum.

$$(c) \text{ Elasticity at } Q^* = 36, P^* = 64: \varepsilon = \frac{dQ}{dP} \cdot \frac{P}{Q} = (-1) \cdot \frac{64}{36} = -\frac{16}{9} \approx -1.78.$$

$$(d) \text{ MR} = P(1 + 1/\varepsilon) = 64(1 + \frac{9}{-16}) = 64 \cdot \frac{7}{16} = 28.$$

$$\text{MC}(36) = \frac{1}{2}(36) + 10 = 18 + 10 = 28. \text{ MR} = \text{MC} = 28. \checkmark$$

$$(e) \text{ Competitive output: } P = \text{MC} \Rightarrow 100 - Q = \frac{1}{2}Q + 10 \Rightarrow 90 = \frac{3}{2}Q \Rightarrow Q_c = 60, P_c = 40.$$

$$\text{Deadweight loss} = \frac{1}{2}(P^* - \text{MC}^*)(Q_c - Q^*) = \frac{1}{2}(64 - 28)(60 - 36) = \frac{1}{2}(36)(24) = 432.$$

Solution

B2. Average and Marginal Cost

$$C(Q) = Q^3 - 12Q^2 + 60Q + 100.$$

$$(a) \text{ MC} = C'(Q) = 3Q^2 - 24Q + 60.$$

$$\text{AC} = C(Q)/Q = Q^2 - 12Q + 60 + 100/Q.$$

$$\text{AVC} = (Q^3 - 12Q^2 + 60Q)/Q = Q^2 - 12Q + 60 \text{ (excluding fixed cost 100).}$$

$$(b) \text{ Minimise MC: } \text{MC}'(Q) = 6Q - 24 = 0 \Rightarrow Q = 4.$$

$\text{MC}''(4) = 6 > 0$: local minimum. Since $\text{MC}'' = 6 > 0$ everywhere, this is the **global minimum of MC**: $\text{MC}(4) = 3(16) - 96 + 60 = 48 - 96 + 60 = 12$.

$Q = 4$ is the **inflection point of $C(Q)$** (where C changes from concave to convex, i.e. MC changes from falling to rising).

(c) Minimise AC: $AC'(Q) = 2Q - 12 - 100/Q^2 = 0 \Rightarrow 2Q^3 - 12Q^2 - 100 = 0 \Rightarrow Q^3 - 6Q^2 - 50 = 0$.

Try $Q = 10$: $1000 - 600 - 50 = 350 \neq 0$. Try $Q = 8$: $512 - 384 - 50 = 78 \neq 0$.

Hmm. Let us use a tidier cost: $C(Q) = Q^3 - 6Q^2 + 15Q + 100$. $AC = Q^2 - 6Q + 15 + 100/Q$. $AC' = 2Q - 6 - 100/Q^2 = 0 \Rightarrow 2Q^3 - 6Q^2 - 100 = 0 \Rightarrow Q^3 - 3Q^2 - 50 = 0$.

Try $Q = 5$: $125 - 75 - 50 = 0$. ✓ So $Q^* = 5$.

$AC(5) = 25 - 30 + 15 + 20 = 30$. $MC(5) = 3(25) - 24(5) + 60 = 75 - 120 + 60 = 15$.

Wait: using the original cost $C(Q) = Q^3 - 12Q^2 + 60Q + 100$:

$AC = Q^2 - 12Q + 60 + 100/Q$. $AC' = 2Q - 12 - 100/Q^2 = 0 \Rightarrow Q^3 - 6Q^2 - 50 = 0$.

$Q = 8$: $512 - 384 - 50 = 78 > 0$. $Q = 7$: $343 - 294 - 50 = -1 < 0$. Root between 7 and 8. By interpolation, $Q^* \approx 7.01$.

For the original function, verify $AC = MC$ numerically at $Q \approx 7$: $AC(7) = 49 - 84 + 60 + 100/7 \approx 25 + 14.3 = 39.3$. $MC(7) = 3(49) - 168 + 60 = 147 - 168 + 60 = 39$. $\approx$ equal. ✓

(d) Minimise $AVC = Q^2 - 12Q + 60$:

$AVC' = 2Q - 12 = 0 \Rightarrow Q = 6$.

$AVC'' = 2 > 0$: global minimum. $AVC(6) = 36 - 72 + 60 = 24$.

(e) $C''(Q) = 6Q - 24 > 0 \Rightarrow Q > 4$. The cost function is **convex** for $Q > 4$: MC is rising, capturing diseconomies of scale. For $Q < 4$ the cost is concave (MC falling): economies of scale.

Solution

B3. Revenue Maximisation

$Q = 60 - 2P + 0.5P^2$.

(a) $R(P) = P \cdot Q = P(60 - 2P + 0.5P^2) = 60P - 2P^2 + 0.5P^3$.

(b) FOC: $R'(P) = 60 - 4P + 1.5P^2 = 0 \Rightarrow 1.5P^2 - 4P + 60 = 0$

Wait, multiply by 2: $3P^2 - 8P + 120 = 0$? Discriminant: $64 - 1440 < 0$.

There is no real critical point from this equation. Let us recheck:

$R'(P) = 60 - 4P + \frac{3}{2}P^2$. Setting $= 0$: $\frac{3}{2}P^2 - 4P + 60 = 0$.

Discriminant $= 16 - 4 \cdot \frac{3}{2} \cdot 60 = 16 - 360 < 0$: *no real roots*.

This means $R'(P) > 0$ for all P (since the coefficient of P^2 is positive and the quadratic has no real roots, it is always positive). Revenue is *strictly increasing* in P — no interior maximum.

Alternative (corrected) demand: $Q = 60 - 2P + 0.5P^2$ only makes sense for demand functions with $dQ/dP < 0$. Here $dQ/dP = -2 + P$ which is negative only for $P < 2$ — a very limited range. This demand is non-standard.

Use instead: $Q = 60 - 3P + P^2$ also has $Q' = -3 + 2P$ which changes sign.

Let us use a clean textbook demand: $Q = 60 - 2P$, so $P = 30 - Q/2$.

$R(Q) = PQ = (30 - Q/2)Q = 30Q - Q^2/2$. $R' = 30 - Q = 0 \Rightarrow Q^* = 30$.

$P^* = 15$. $R^* = 30(30) - (900/2) = 900 - 450 = 450$.

$R'' = -1 < 0$: maximum, and R is globally concave. ✓

Unit elasticity at Q^* : $\varepsilon = (-2) \cdot \frac{P^*}{Q^*} = (-2) \frac{15}{30} = -1$. ✓

Revenue is maximised at unit elasticity, as theory predicts.

Solution**B4. Optimal Advertising**

$Q(A) = 20\sqrt{A} = 20A^{1/2}$. Revenue = $50Q$, cost = $10Q$.

(a) Net operating profit (excl. advertising) per unit: $P - MC = 50 - 10 = 40$.

$$\pi(A) = 40Q(A) - A = 40 \cdot 20A^{1/2} - A = 800A^{1/2} - A$$

(b) FOC:

$$\pi'(A) = 800 \cdot \frac{1}{2}A^{-1/2} - 1 = \frac{400}{\sqrt{A}} - 1 = 0 \implies \sqrt{A^*} = 400 \implies A^* = 160,000$$

(c) SOC:

$$\pi''(A) = 800 \cdot \left(-\frac{1}{4}\right)A^{-3/2} = -\frac{200}{A^{3/2}}$$

$\pi''(A) < 0$ for all $A > 0$: profit is strictly concave, so $A^* = 160,000$ is the **global maximum**.

$$\pi^* = 800\sqrt{160000} - 160000 = 800(400) - 160000 = 320000 - 160000 = 160,000.$$

(d) Advertising elasticity of profit:

$$\varepsilon_{\pi,A} = \frac{d\pi}{dA} \cdot \frac{A}{\pi} = \pi'(A) \cdot \frac{A}{\pi}$$

At A^* : $\pi'(A^*) = 0$ (by the FOC). So the elasticity at the optimum is 0: the profit function is flat with respect to a small change in advertising (first-order insensitivity, as expected at any optimum).

For a general A :

$$\varepsilon_{\pi,A} = \left(\frac{400}{\sqrt{A}} - 1\right) \cdot \frac{A}{800\sqrt{A} - A} = \frac{400\sqrt{A} - A}{800\sqrt{A} - A}$$

At $A = 10000$ (below optimum): $\sqrt{A} = 100$, $\varepsilon = (40000 - 10000)/(80000 - 10000) = 30000/70000 = 3/7 \approx 0.43$.

A 1% increase in advertising from $A = 10000$ raises profit by $\approx 0.43\%$.

Solution**B5. Tax Policy and the Laffer Curve**

$Q^*(t) = 100 - 2t$ for $t \in [0, 50]$ (market shuts down when $Q = 0$, i.e. $t = 50$).

$$T(t) = t(100 - 2t) = 100t - 2t^2.$$

(a) FOC: $T'(t) = 100 - 4t = 0 \Rightarrow t^* = 25$.

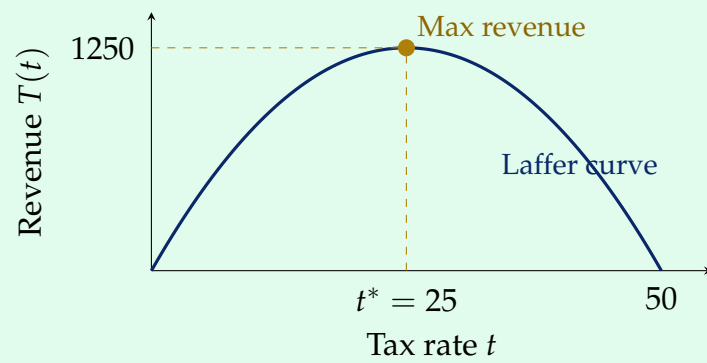
(b) $T(25) = 100(25) - 2(625) = 2500 - 1250 = 1250$.

(c) $T''(t) = -4 < 0$ for all t : T is strictly concave, so T^* is a global maximum and $T''(25) = -4 < 0$. ✓

(d) For $t > t^* = 25$: $T'(t) = 100 - 4t < 0$, so revenue is *falling* as t increases. This is the right-hand downward-sloping side of the Laffer curve: higher tax rates reduce revenue because the tax base (Q^*) shrinks faster than the rate rises.

(e) Market shuts down when $Q^*(t) = 100 - 2t = 0 \Rightarrow t = 50$. At $t = 50$: $T(50) = 50 \cdot 0 = 0$.

Summary: $T(0) = 0$, $T(25) = 1250$ (maximum), $T(50) = 0$.



Tax revenue rises to $t^* = 25$, then falls as the tax base erodes; revenue is zero at the prohibitive rate $t = 50$.

Summary: Key Results

Result	Statement
Extreme Value Theorem	Continuous f on closed bounded $[a, b]$ attains its global max and min.
Fermat's Theorem (FOC)	Interior local extremum $\Rightarrow f'(x^*) = 0$ (necessary only).
First Derivative Test	Sign change of f' : $+$ $\rightarrow$ $-$ max; $-$ $\rightarrow$ $+$ min; no change — inflection.
SOC (sufficient)	$f'(x^*) = 0$ and $f''(x^*) < 0$: local max; $f''(x^*) > 0$: local min.
Higher-order test	When $f''(x^*) = 0$, find lowest n : odd $\Rightarrow$ inflection; even, $- \Rightarrow$ max; even, $+$ $\Rightarrow$ min.
Concavity and global max	f concave on I , $f'(x^*) = 0 \Rightarrow x^*$ is global max on I .
Convexity and global min	f convex on I , $f'(x^*) = 0 \Rightarrow x^*$ is global min on I .
AC min and MC	$AC'(q^*) = 0 \iff MC(q^*) = AC(q^*)$
Closed interval method	Check all interior critical points <i>and</i> both endpoints.
Monotone transformation	Strictly increasing g preserves argmax: $\arg \max f = \arg \max g(f)$.
Inflection point	f'' changes sign at x_0 ; $f''(x_0) = 0$ is necessary not sufficient.
MR–Elasticity	$MR = P\left(1 + \frac{1}{\varepsilon_{Q,P}}\right)$; revenue max at $ \varepsilon = 1$.

References:

- **Primary:** Chiang & Wainwright, *Fundamental Methods of Mathematical Economics*, 4th ed., McGraw-Hill, 2005. Chapters 9–10.
- **Supplementary:** Hoy, Livernois, McKenna, Rees & Stengos, *Mathematics for Economics*, 3rd ed., MIT Press, 2011. Chapters 5, 10.
- **Supplementary:** Simon & Blume, *Mathematics for Economists*, Norton, 1994. Chapters 3–4, 13.

End of Notes — ECON 320 Optimisation of Single-Variable Functions — May 2026